



Adapter for High Pressure Fuel Pump Cleaning

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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February 2022
4.03-220201-1

1.0 Introduction

This improvement describes how to perform the cleaning process in the lubrication line for high pressure fuel pumps, during the process of C175 engine partial overhaul, without failure by utilizing an adapter tool that Ferreyros designed.

2.0 Best Practice Description

The improvement consists of using an adapter on the high-pressure fuel pump of C175 engines and connecting it to the upper oil inlet line, using the oil-supplied injection pump test stand to perform the cleaning and remove the remaining used oil.

The following steps are required to use this tool. Refer to Section 6.0 for images.

1. Install the high-pressure pump lifting bracket (P/N 487-8643).
2. Install the high-pressure pump on the injection pump test stand.
3. Install the adapter on the top oil line of the high-pressure pump.
4. Install the test stand oil supply hose onto the adapter.
5. Turn on the test stand.
6. Regulate the oil supply pressure between 45 to 50 PSI.
7. Flush with clean oil for approximately 30 minutes.

3.0 Implementation Steps

Design the adapter to be used for connection to the lubrication line of the high-pressure fuel pump. The adapter is shown below and should be designed according to the blueprint attached in Section 6.0.



Figure 1: Multipurpose tool for assembling the C175 engine auxiliary water pump.

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4.0 Benefits

- Prevents contamination in the process of repairing the engine by removing remaining oils that may contain contaminants inside the lubrication system of the high-pressure fuel pump;
- Avoids related failures due to contamination.

5.0 Resources Required

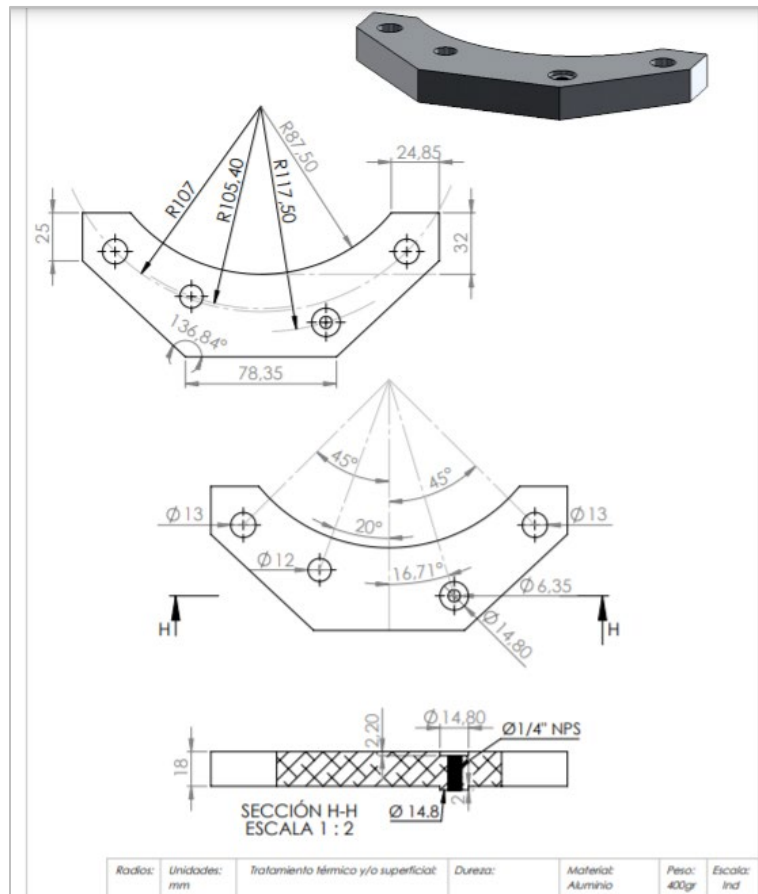
Item	Cost
Adapter	\$42

6.0 Supporting Attachments / References

M0072382 (Inspection, Salvage, Reuse, and Cleaning of the C175 Tier 2 Engine Fuel Rails and Critical Fuel Lines)

M0122417 (Inspection, Salvage, Reuse, and Cleaning of 3500 Tier 4 and C175 Tier 4 Engines Fuel Rails and Critical Fuel Lines)

Engineering print for the fabrication of the adapter:



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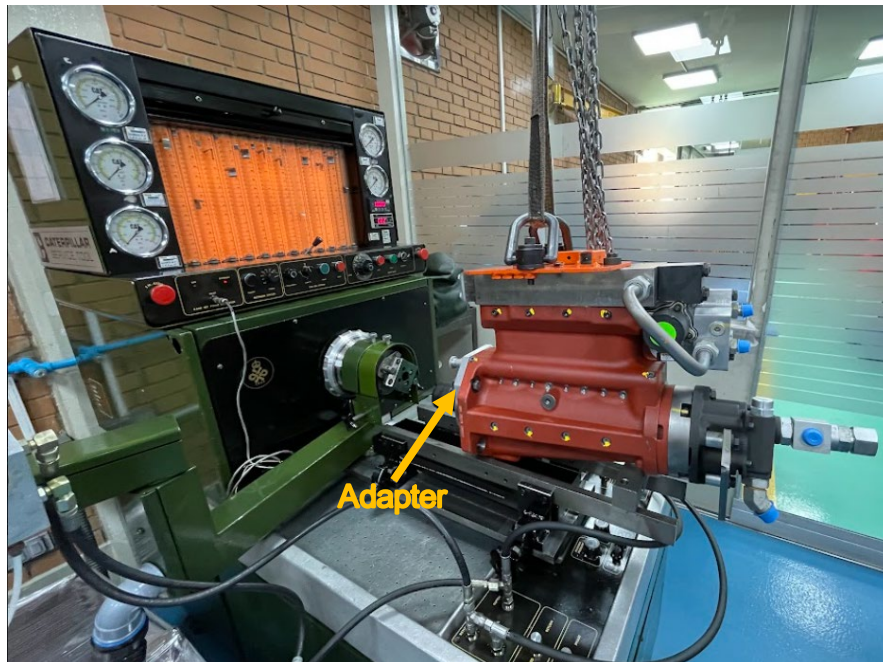
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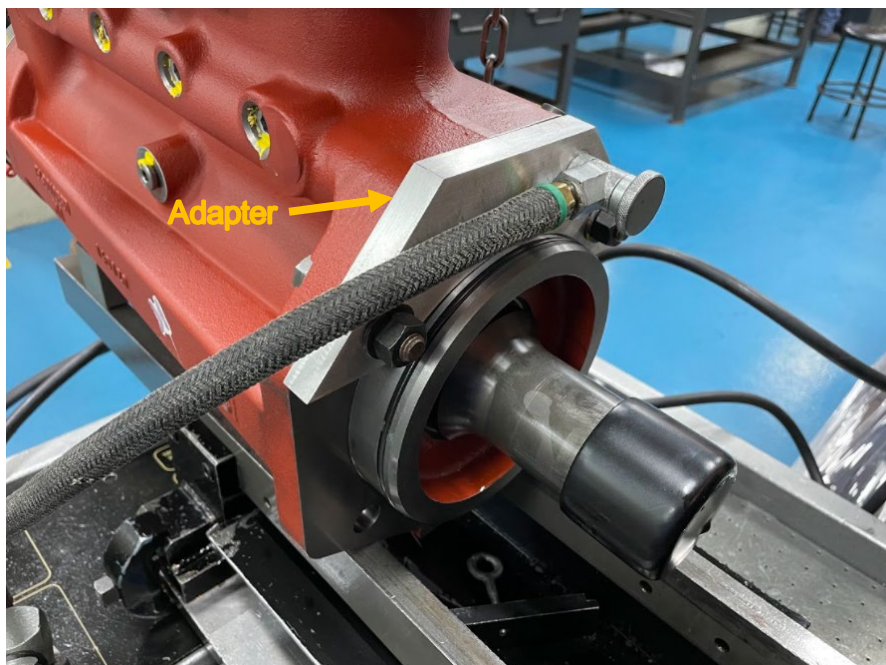
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- Installation of the high-pressure fuel pump on the injection pump test stand using bracket P/N 487-8643.



- Installation of the adapter and oil supply hose:



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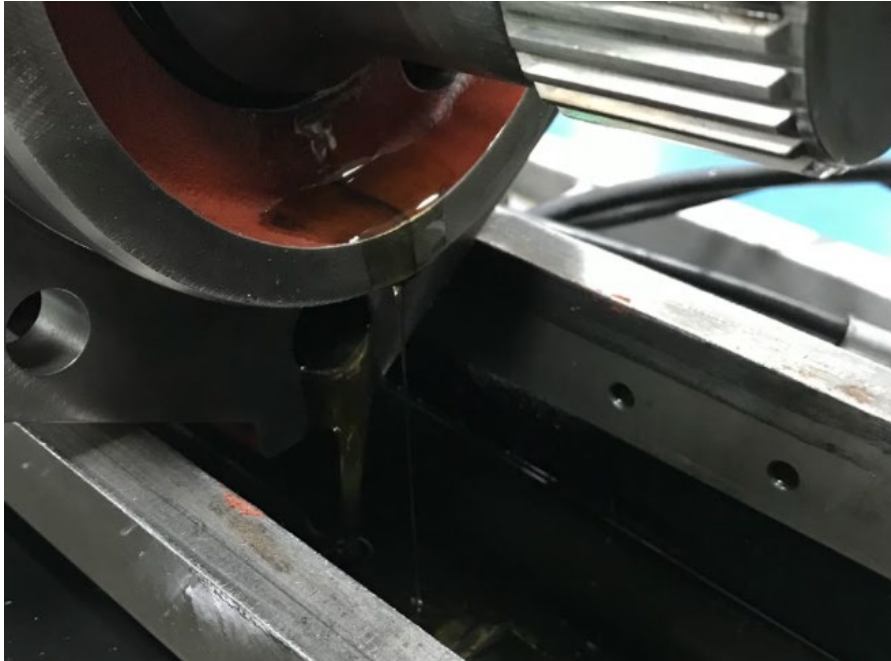
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- Supply oil from the injection pump test bench with a pressure between 45 to 50 PSI to perform the cleaning:



7.0 Related Best Practices

N/A.

8.0 Acknowledgements

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